Database Development and Implementation

(M31994 and M32275)

Coursework

(Referral & Deferral coursework)

2025

Learning Outcomes:

1. Develop an understanding of data management concepts and evaluate its theories and techniques particularly about Big Data.
2. Explain the theoretical underpinning of the relational model and apply that to developing a relational database from given business user requirements.
3. Manipulate real life datasets using industry standard software to produce management information in an easily assimilated format.

Part 1: 'Happy Eat' Restaurant Database Management



A new restaurant named 'Happy Eat' is scheduled to open in the Portsmouth commercial area at the end of 2024. It will accept customer orders from 6PM to 11PM, Monday through Saturday.

Each order can include one or more dishes, and each dish is prepared by one chef. Each chef can prepare multiple dishes. Each order can include multiple dishes, and each dish can be part of multiple orders. An order might include multiple same dishes (e.g., pizza x 2). Each dish belongs to a certain category (e.g., starters, mains, desserts) and is prepared by a chef. Each chef can prepare multiple dishes. Additionally, each chef works in a specific kitchen station designated for the preparation of certain categories of dishes. The restaurant also tracks the ingredients used for each dish.

The restaurant also can take online orders. Each order can be delivered by one driver, and one driver can take multiple orders.

You are required to design a relational database for Happy East restaurant to enable efficient information management and enhanced customer service. The necessary information they require include: customer contact details; the orders and their respective dishes along with their categories; chef details and their assigned kitchen stations, ingredients for each dish and driver information for delivery each order.

Required:

After carefully reading and analysing the case study you are required to carry out the following tasks:

i. Create a fully labelled and implementable Crow's Foot ERD based on your analysis of above scenario. Include all business rules, entities, keys, relationships, and cardinalities. **(20%)**

- **State fully the assumptions you have made in constructing the ERD.**
- **Do include key attributes (FK and FK) and non-key attributes (**a listing of all the attributes that in your opinion are required to produce the information specified in the scenario) **in your diagram.**

NOTE: You are advised to use pencil and paper until you have reached a satisfactory model and then re-create this in Lucidchart. Visit this link for access to, and a tutorial on, this software package: http://holowczak.com/drawing-entity-relationship-diagrams-with-crows-foot-notation-using-lucidchart/

ii. Write a **500-word** paragraph, **in non-technical terms,** that provides 'Happy Eat' management of some feedback (related to one aspect of their services) about how you, as a customer, hope to improve your experiences. (**10 marks).**

iii. Based on your first ERD answer, discuss how you, as a database designer, collect and add that information (feedback from your answer in ii part) to this existing database to enhance customer service or information management. You DO need to produce a new ERD for the potential added information using Lucidechart. **(20 marks)**

Part 2: SQL practice

Suppose you are working as a data analyst for McDonald's European division. Your manager has some questions regarding the restaurant's operations. Please use the three provided tables and SQL query codes to extract the relevant information and answer your manager's questions.

Here is some information about the three files (tables):

Sales: This dataset contains information about specific transactions, such as the product names, prices, quantities, costs, purchase types and payment methods.
Store: This dataset provides information about McDonald's stores, including Store ID, country, region, general manager's name, age, sex and tenure.
Theft: This dataset contains information about the stolen transactions.

Question 1:

1.1 What types of products are we currently selling? **(7 marks)**
1.2 What is our average price for each product? **(10 marks)**

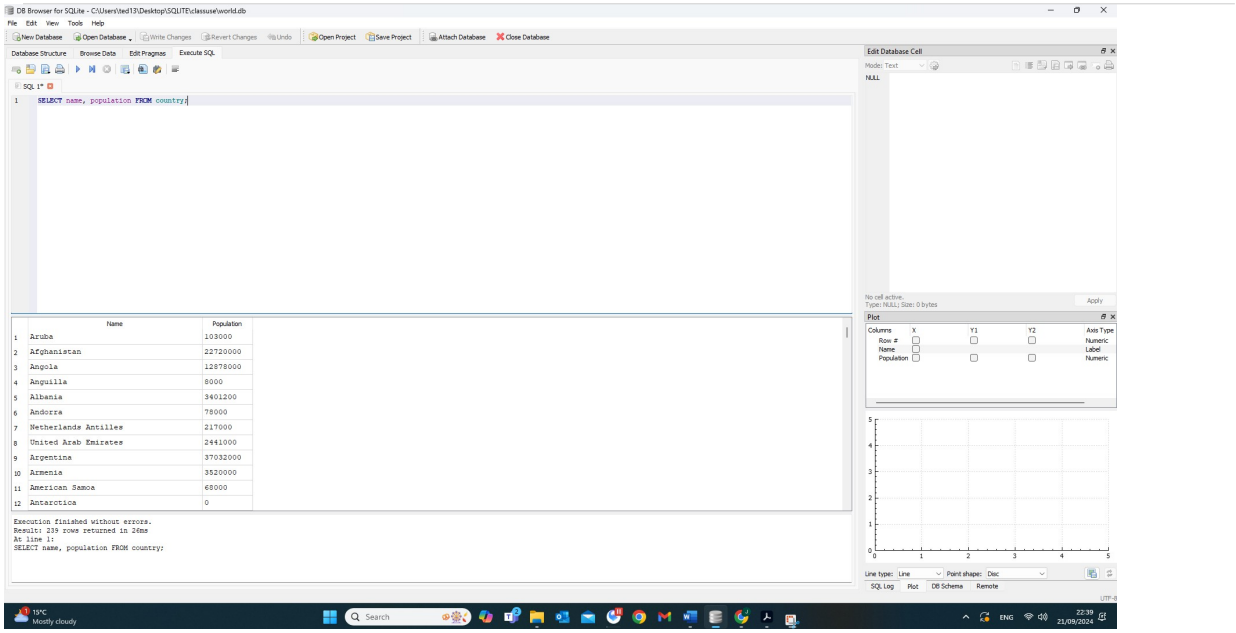
Question 2:

2.1 How much does each manager have in theft amounts? **(10 marks)**
2.2 Filter by the managers that have more than 10 in theft amounts and find their age, sex, tenure, and country_city location. **(5 marks)**

Question 3:

3.1 Assuming the data is clean, use SQL to determine which city has the highest sales (in terms of quantity) and what the highest sales are. **(8 marks)**
3.3 Assuming the data is clean, list all the products by quantity sold from high to low and show their quantities. **(10 marks)**

Note: Please copy our screenshots which have your SQL code, output and process time information to your word file and submit your work as a PDF file. (For example, the following screen is what is expected. In the picture, the actual code and result serve only as an example.)



Submission requirements:

- **Deadline**: 13:00, Monday, 14th July 2025
- Word count: 1,500 words maximum (word count from EDR and assumption statements in task i is counted)
- Submission format: PDF file

(to be continued)

Submission Hand in Arrangements: Students are required to submit their coursework assignment to **WiseFlow** via the link on the module Moodle. The submission dropbox will show around one week before the submission deadline. Students are encouraged to submit drafts through the **WiseFlow** student checkpoints prior to the submission deadline.

There are mark penalties for late hand in. (Corruption of computer files is not an adequate excuse for late hand in, as work should be adequately backed up.)

PLEASE NOTE:

This assessment is eligible for the automatic 48-hour extension.

If you're facing short-term circumstances that mean you're not able to submit your assessment by the published deadline, you can make use of the Automatic Extension. This gives you a further 48 hours to submit your work - and you don't have to do anything to request the extension.

The end of the Automatic Extension period will be 1pm on Thursday 19 December.

Please note, you won't have access to any academic support relating to completing your assessment during the 48-hour extension period.

Up until the point of the original deadline, you may overwrite your submission. Once the original deadline has passed, you may not resubmit your work even if it is within the automatic 48-hour extension period. If you need to make use of the automatic 48-hour extension, you may only submit your work once during the 48-hour period.

However, students are encouraged to submit in advance of the final deadline wherever possible. Computer uploading delays and other IT difficulties are not accepted as extenuating circumstances.

If there is a problem with Moodle at the time of submission students should email their coursework to submissions@port.ac.uk with the following details:

Student Number

Module Name / Module number

Course name that you are studying

Screenshot of the error

Word Limit: **1000** words excluding bibliography. The word count should be stated on the front page of the document. Failure to state a word count will result in a penalty of 5% of the original mark awarded. A falsely stated word-count is an assessment office which may result in a penalty, including the reduction of the mark to 0%. Examiners are free to disregard footnotes that contain inappropriate information or information that should belong in the main text. Coursework that is over the stated word limit will result in a penalty of 10% of the original mark awarded**. For the avoidance of doubt, the penalty will be applied to any work that exceeds the stated word limit of 1000 words excluding bibliography. Students are NOT permitted to exceed the word limit by 10% or any other amount.**

Referencing requirements: Students must reference all sources using the APA 7th Edition. Guidance on this method of referencing can be found at www.referencing.port.ac.uk. Reference should be made to the primary source, except when the primary source can no longer be obtained. Poor citation of sources will result in a loss of marks.

Referencing is required to give intellectual credit to your source, help your reader recover your source easily and to avoid being accused of plagiarism.

Students are reminded that the University will not tolerate academic dishonesty in any form. This is cheating.

Plagiarism: Students are reminded of the need to avoid plagiarism. The University Regulations describe plagiarism as:

> the incorporation by a student in work for assessment of material which is not their own, in the sense that all or a substantial part of the work has been copied without any adequate attempt at attribution, or has been incorporated as if it were the student's own when in fact it is wholly or substantially the work of another person or persons.

Any student suspected of plagiarising will be referred for an Academic Misconduct Hearing. Please ensure that the final product is your own work, and not just copied from an AI generator, in whole or in part. Generated text or output may be used as a starting point to give inspiration or guidance, but the final submitted assessment must all be your own work. See further guidance at https://library.port.ac.uk/w211/82
Students should ensure that all sources are cited in the bibliography and that indentation or quotation marks (as appropriate) are used when quoting. Failure to include a bibliography will result in a 5% penalty, unless the lecturer/tutor has advised you that a bibliography is not required.

Formatting: The work should be word processed. Font size should be between 11 and 14 and 'easy to read' e.g. Calibri, Arial, Times New Roman. The Header must include the student number and the Footer must include a page number.

Electronic Copy of Work: Students should retain an electronic copy of their coursework, so that it may be checked by a member of staff should a member of staff feel the need to do so. Tutors are entitled to request an electronic copy of coursework if they have any doubt about the accuracy of the stated word count and/or any suspicion of plagiarism. Failure to send an electronic copy of the coursework to a member of staff who has asked for a copy may result in a penalty.
If any student has a query about any of the above matters and wishes to obtain clarification or further information, please contact your Module Co-ordinator or personal tutor.

Feedback: Marking will be done in accordance with the grade descriptors in the table below and the categorical marking criteria for the University of Portsmouth for PG level 7 as set out in Table 6 of the Assessment Regulations for Undergraduate and Postgraduate Courses.

http://www2.port.ac.uk/departments/services/dcqe/storeddocuments/filetodownload,171073,en.pdf

	Levels 7 & 8
80+	As below plus: · Excellent work – able to express an original reasoned argument in a lucid manner by reviewing and critiquing a wide range of material. Original, critical thinking based on outstanding insight, knowledge and understanding of material. Material contributes to current understanding and is of potentially publishable quality in terms of presentation and content. · Wide reaching research showing breadth and depth of sources.
70–79	
60–69	As below plus: · Clear, balanced coherent critical and rigorous analysis of the subject matter. Detailed understanding of knowledge and theory expressed with clarity. · Extensive use of relevant and current literature to view topic in perspective, analyse context and develop new explanations and theories.

50–59	As below plus: · Detailed review and grasp of pertinent issues and a critical contextual overview of the literature. Thorough knowledge of theory and methods and uses this to underpin arguments and conclusions. · Confidence in understanding and using literature.
40–49	· Demonstrates grasp of key concepts and an ability to develop and support an argument in a predominately descriptive way with valid conclusions drawn from the research. · Familiarity with key literature which is cited and presented according to convention. · Logical and clear structure, well organised with good use of language and supporting material.
0–39	FAIL – Some knowledge of relevant concepts and literature but significant gaps in understanding and / or knowledge. Little attempt at evaluation, conclusions vague, ambiguous and not based on researched material. Limited or inappropriate research. Deficits in length, structure, presentation and / or prose.